

Master Study Guide: Construction Materials (CE 126)

Unit 1: Introduction & Engineering Properties of Materials — [6.5% - 6.5%] of Total Marks

A. Short Answer Questions (SAOs)

2081 Q1 (8 Marks):

How is civil engineering materials classified? Define the terms: Porosity, Toughness, Creep, Brittleness, fatigue and resilience. [2+6]

2080 Q1 (8 Marks):

As a civil engineer, what are the major factors which you consider most for the selection of material in civil engineering project in Nepal? Write short notes on significant mechanical properties of civil engineering materials. [8]

2079 Q1 (8 Marks):

Define the scope of construction materials and their types based on the physical and chemical characteristics. (3+5)

B. Long Answer Questions (LAQs)

No long answer questions have been asked directly from this introductory unit in the end-semester examinations.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This unit consistently appears as Q1 in Group A of the final end-semester examinations, carrying exactly 8 marks. The questions are purely theoretical, requiring clear definitions, classification lists, and explanations of fundamental material properties or factors influencing material selection in Nepal.*
- **Direct Repetitions:**
 - *"Significant mechanical/engineering properties of materials" (asked generally in 2080 Q1, and with specific terms like Porosity, Creep, and Toughness in 2081 Q1).*
 - *"Classification of construction materials" (asked by physical/chemical characteristics in 2079 Q1, and general classification in 2081 Q1).*
- **Key High-Priority Topics:**

- *Fundamental engineering properties: Porosity, Toughness, Creep, Brittleness, Fatigue, Resilience, Elasticity, and Plasticity.*
 - *Multi-criteria decision-making for materials selection in Nepal's diverse geography (accessibility, durability, cost, local availability).*
 - *Scope and categorization of materials (physical, chemical, and geological classification).*
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Unit 2: Building Stones — [8.9% - 10.6%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Q1 (5 Marks):

Write in detail about the various classifications of stones. [5]

2081 Q11.a (5 Marks):

Test for stones. [5] (*Short note*)

2081 Q7.a (2 Marks):

Give in brief specifications for good building stone. [2] (*Part of a long question*)

2081 Q2.a (4 Marks):

Write down the purposes of dressing of stones. What are the methods to preserve the stone from deterioration? [2+2]

2079 Q10.a (5 Marks):

What requirements are important for a good location of quarrying stones to be fulfilled? [5] (*Part of a long question*)

2079 Q5 (8 Marks):

Why tests are crucial for building stones? Define the procedure of crushing test and water absorption test of stone. [2+3+3]

B. Long Answer Questions (LAQs)

2080 Q8.a (12 Marks):

Explain the various types of tests on building stones and provide briefly specifications for good building stone. [12]

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Focuses on both theoretical requirements (such as dressing, quarrying, and specifications) and practical laboratory testing procedures. Long questions (12 marks in 2080 Q8.a) require detailed step-by-step testing procedures.
- **Direct Repetitions:**
 - **REPEATED: Similar Concept** — "Specifications for good building stone" was asked in 2081 Q7.a (2 marks) and 2080 Q8.a (part of 12 marks).
 - **REPEATED: Similar Concept** — "Tests for building stones" was asked as a major descriptive question in 2080 Q8.a (12 marks), 2079 Q5 (8 marks), and as a short note in 2081 Q11.a (5 marks).
- **Key High-Priority Topics:**
 - Laboratory testing of stones: **Crushing Strength test, Water Absorption test, Durability test, and Acid test.**
 - Methods of stone preservation from weathering and acid rain.
 - Purpose and methods of dressing of stones (rough tooling, hammer dressing, chisel drafting).
 - Selection criteria and geological/physical/chemical classification of stones.

Unit 3: Clay Products (Bricks and Tiles) — [9.8% - 13.8%] of Total Marks

A. Short Answer Questions (SAOs)

2082 Q5.b (4 Marks):

Write about the Constituents of brick. [4] *(Part of a 2082 Mid-Term Q5)*

2082 Q2 (5 Marks):

What are the tests for the acceptance of bricks for building construction? Describe the characteristics of good bricks. [3+2]

2081 Q10.a (3 Marks):

Define efflorescence in bricks. How can it be removed? [3]

2081 Q4.b (4 Marks):

Describe about the tiles and their types. [4]

2079 Q9.b (6 Marks):

Describe in brief about roof and floor tiles. [6] *(Part of a long question)*

2079 Q4 (8 Marks):

Suppose your neighbor demands a technical assistance from you regarding selection of a good bricks. At this moment, what must be your role for the selection of good brick and its defects? [8]

B. Long Answer Questions (LAQs)

2081 Q7.a (10 Marks):

Explain the various types of tests on bricks. [10] (*Part of a long question*)

2080 Q7.a (12 Marks):

With neat sketches, explain in detail about the manufacturing process of conventional bricks. [12]

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Questions are a mix of technical testing procedures, field checks, chemical compositions, and manufacturing flowcharts/sketches.
- **Direct Repetitions:**
 - **REPEATED: Similar Concept** — "Tests on bricks" is repeated in 2082 Q2 (3 marks) and 2081 Q7.a (10 marks).
 - **REPEATED: Near Identical** — "Tiles and their types / Roof and floor tiles" is asked in 2081 Q4.b (4 marks) and 2079 Q9.b (6 marks).
 - **REPEATED: Similar Concept** — "Characteristics and selection criteria of good bricks" is repeated in 2082 Q2 (2 marks) and 2079 Q4 (8 marks).
- **Key High-Priority Topics:**
 - Step-by-step brick manufacturing process: **Preparation of clay (unsoiling, digging, weathering, blending, tempering), Moulding (hand vs. machine), Drying, and Burning (Clamps vs. Kilns).**
 - Essential brick tests: **Compressive strength, Water absorption, Efflorescence, Hardness, Soundness, and Structure.**
 - Brick defects: **Efflorescence (causes, prevention, and removal), Bloating, Chuffs, and Lamination.**
 - Classification and engineering applications of tiles (roofing, flooring, glazed wall tiles).

Unit 4: Lime & Lime Mortar — [3.3% - 7.7%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Q5.a (4 Marks):

Comparison between different types of lime. [4] (*Part of a 2082 Mid-Term Q5*)

2081 Q5 (8 Marks):

Differentiate quick lime with hydraulic lime. What do you mean by calcination and slaking of lime? [4+4]

2080 Q7.b (3 Marks):

Differentiate between lime and cement. [3] (*Part of a long question*)

2080 Q3 (8 Marks):

What is lime cycle? Explain manufacturing process of lime in detail. [8]

2079 Q3.b (4 Marks):

What are the properties of lime mortars? [4] (*Part of a short question*)

B. Long Answer Questions (LAQs)

No standalone long answer questions have been asked directly from this unit.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *Purely descriptive and chemical comparison questions. Focuses heavily on the chemical changes occurring during lime processing.*
- **Direct Repetitions:**
 - **REPEATED: Similar Concept** — "Quick lime vs. Hydraulic lime differentiation / Comparison of different types of lime" is asked in 2081 Q5 (4 marks) and 2082 Q5.a (4 marks).
- **Key High-Priority Topics:**
 - **The Lime Cycle:** Complete chemical sequence from Calcium Carbonate (Limestone, CaCO_3) → Calcium Oxide (Quicklime, CaO) → Calcium Hydroxide (Slaked/Hydrated Lime, Ca(OH)_2) → Calcium Carbonate (CaCO_3).
 - Distinction between Quick Lime, Fat Lime, and Hydraulic Lime (based on clay content and setting properties).
 - Chemistry of slaking (hydration) and calcination

Unit 5: Cement — [5.7% - 12.6%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Q3 (5 Marks):

Describe the Soundness and Compressive Strength test for Cement in detail. [5]

2081 Q7.b (3 Marks):

Write down the reaction in the kiln during manufacture of the cement. [3] *(Part of a long question)*

2081 Q4.a (4 Marks):

Explain in detail about various constituents of cement. [4] *(Part of a short question)*

2080 Q11.a (5 Marks):

Bouge's compounds. [5] *(Short note)*

2080 Q10.a (6 Marks):

Write down the reaction in the kiln during manufacture of the cement. [6] *(Part of a long question)*

2080 Q8.b (3 Marks):

Differentiate between setting and hardening of cement. [3] *(Part of a long question)*

2079 Q9.a (4 Marks):

Define the dry process while manufacturing OPC. [4] *(Part of a long question)*

2079 Q8.b (7 Marks):

What are the main properties of the compounds of clinker? [7] *(Part of a long question)*

2079 Q6.a (4 Marks):

Define high alumina cement and rapid hardening cement. [4] *(Part of a short question)*

B. Long Answer Questions (LAQs)

2082 Q4 (8 Marks):

Explain with flow chart the step-by-step dry process of manufacturing of cement. [8] *(2082 Mid-Term LAQ)*

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This unit represents a massive block of marks. Questions focus heavily on industrial manufacturing chemical pathways, clinker chemistry, and laboratory testing procedures.
- **Direct Repetitions:**
 - **REPEATED: Identical** — "Reaction in the rotary kiln during cement manufacture" was asked in 2081 Q7.b (3 marks) and 2080 Q10.a (6 marks).
 - **REPEATED: Near Identical** — "Bogue's (Bouge's) Compounds / Clinker Compounds and their properties" was asked in 2080 Q11.a (5 marks) and 2079 Q8.b (7 marks).
 - **REPEATED: Similar Concept** — "OPC dry manufacturing process" was asked in 2082 Q4 (8 marks) and 2079 Q9.a (4 marks).
- **Key High-Priority Topics:**
 - **Bogue's Compounds:** Tricalcium Silicate (C_3S - Alite), Dicalcium Silicate (C_2S - Belite), Tricalcium Aluminate (C_3A - Celite), Tetracalcium Aluminoferrite (C_4AF - Felite) — their rates of hydration, heat generated, and contribution to strength.
 - Chemical reactions in different zones of the rotary kiln (Drying, Pre-heating, Calcining, Clinkering/Burning, and Cooling zones).
 - **Dry process flowchart:** Crushing → Proportioning → Grinding → Pre-heating → Kiln burning → Clinker cooling → Gypsum addition → Final grinding.
 - Essential cement tests: **Soundness (Le Chatelier/Autoclave), Compressive Strength (mortar cubes), Consistency (Vicat apparatus), and Fineness.**
 - Specific cements: **Rapid Hardening Cement, High Alumina Cement, Portland Pozzolana Cement (PPC).**

Unit 6: Concrete & Admixtures — [16.3% - 22.0%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Q6 (4 Marks):

Distinguish between bleeding and segregation. [4]

2081 Q11.c (5 Marks):

Admixtures. [5] (*Short note*)

2080 Q11.b (5 Marks):

Mineral admixtures. [5] (*Short note*)

2079 Q6.b (4 Marks):

What are the main functions of admixtures? [4] (*Part of a short question*)

2079 Q2 (8 Marks):

Why are joints in concrete structures important? Explain their types. [2+6]

B. Long Answer Questions (LAQs)

2081 Q9.a (12 Marks):

List the properties of fresh and hardened concrete. With sketches explain any two tests for testing the workability of self-compacting concrete. [12] (*Part of a long question*)

2081 Q8.a (10 Marks):

Describe in detail about the manufacturing process of concrete in detail. [10] (*Part of a long question*)

2080 Q9 (15 Marks):

List the fresh and hardened properties of concrete. Explain any two for each fresh and hardened properties of the concrete with neat sketch. [15]

2079 Q10.b (10 Marks):

Define curing, its method and effects of improper curing. [10] (*Part of a long question*)

2079 Q7.a (5 Marks):

What are the main conditions favorable for segregation? [5] (*Part of a long question*)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This is the highest-weight unit in the syllabus, commanding around 20% of the entire final exam marks. It demands heavy technical depth, step-by-step descriptions of concrete production stages, curing protocols, workability testing, concrete defects, and admixture classification.*
- **Direct Repetitions:**
 - **REPEATED: Near Identical** — *"Fresh and hardened properties of concrete with neat sketches of tests" was asked in 2081 Q9.a (12 marks) and 2080 Q9 (15 marks).*
 - **REPEATED: Similar Concept** — *"Segregation and Bleeding (definitions, causes, prevention, and comparison)" was asked in 2082 Q6 (4 marks) and 2079 Q7.a (5 marks).*
 - **REPEATED: Highly Repeated Topic** — *"Admixtures, their functions, and types" was asked in 2081 Q11.c (5 marks), 2080 Q11.b (5 marks), and 2079 Q6.b (4 marks).*
- **Key High-Priority Topics:**
 - **Concrete Production Steps:** *Batching (volume vs. weight), Mixing (hand vs. machine), Transporting, Placing, Compacting, Curing, and Finishing.*
 - **Workability and its Tests:** *Slump test, Compacting Factor test, Vee-Bee consistometer test, and specialized self-compacting concrete tests (L-Box, V-Funnel, Slump Flow).*
 - **Curing of Concrete:** *Hydration heat management, methods (ponding, spraying, steam curing, membrane curing), and consequences of improper curing (cracking, low durability, strength loss).*
 - **Concrete Joints:** *Construction, expansion, contraction, and isolation joints—their purposes and locations.*
 - **Admixtures:** *Chemical (plasticizers, superplasticizers, retarders, accelerators, air-entraining agents) and Mineral (fly ash, silica fume, GGBS).*

Unit 7: Timber & Wood Products — [6.5% - 10.6%] of Total Marks

A. Short Answer Questions (SAQs)

2081 Q10.c (4 Marks):

Write about the products of timber & their uses. [4] *(Part of a long question)*

2081 Q3 (8 Marks):

What is seasoning of timber? List out and explain with neat sketch the various defects found in timber. [8]

2080 Q2 (8 Marks):

List out and explain with neat sketch the various defects found in timber. [8]

2079 Q3.a (4 Marks):

What types of diseases you may find in wood? Explain them in brief. [4] *(Part of a short question)*

B. Long Answer Questions (LAQs)

2079 Q11.a (9 Marks):

Enumerate the different methods of preservation of timber. [9] *(Part of a long question)*

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Timber is a regular chapter in both Group A and Group B. It leans heavily on botanical and physical defects/diseases, timber treatment engineering (seasoning and preservation), and industrial wood derivatives.
- **Direct Repetitions:**
 - **REPEATED: Identical** — "Defects found in timber with neat sketches" is 100% repeated in 2081 Q3 (8 marks) and 2080 Q2 (8 marks).
- **Key High-Priority Topics:**
 - **Timber Defects:** Natural/growth defects (shakes - heart, star, cup, radial; knots - dead, live; wind cracks; rind gall) and conversion/seasoning defects (warping, bowing, twisting, cupping).
 - **Timber Seasoning:** Natural seasoning (air, water) and Artificial seasoning (kiln, chemical, electrical, boiling) — objectives and process details.

- **Timber Preservation:** Chemical treatments (creosote oil, ASCU treatment, coal tar, organic solvent preservatives) and application methods (brushing, spraying, pressure impregnation).
 - **Timber Diseases:** Dry rot, Wet rot, and insect attacks (termites, beetles).
 - **Timber Products:** Plywood, Laminates, Fiberboards, Particle boards, Veneers—composition and applications.
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Unit 8: Metals & Alloys — [9.8% - 10.6%] of Total Marks

A. Short Answer Questions (SAQs)

2081 Q11.b (5 Marks):

Steel Alloys [5] (*Short note*)

2081 Q10.b (4 Marks):

What are the various factors affecting the properties of steel? Explain. [4] (*Part of a long question*)

2081 Q9.b (3 Marks):

Enumerate the advantages and disadvantages of using aluminum as building material. [3] (*Part of a long question*)

2080 Q10.b (4 Marks):

What is steel? What are the reasons for the manufacture of steel? How can you identify the steel and wrought iron? [4] (*Part of a long question*)

2080 Q4 (8 Marks):

Describe in short about different iron ores. [8]

2079 Q8.a (3 Marks):

Describe the manufacturing process of formation of cast iron. [3] (*Part of a long question*)

B. Long Answer Questions (LAQs)

2079 Q7.b (10 Marks):

Define steel and its properties. What are the factors affecting the properties of steel? [10] (*Part of a long question*)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Questions are highly structured around metallurgy, structural steel characteristics, comparative material analysis (steel vs. wrought iron, aluminum advantages), and raw mining materials (iron ores).
- **Direct Repetitions:**
 - **REPEATED: Identical** — "Factors affecting the properties of steel" was asked in 2081 Q10.b (4 marks) and 2079 Q7.b (part of 10 marks).
- **Key High-Priority Topics:**
 - **Steel Properties & Factors:** Carbon content (mild steel vs. high carbon steel), presence of impurities (sulfur, phosphorus), alloying elements, and heat treatment (annealing, quenching, tempering).
 - **Iron Metallurgy:** Major iron ores (Hematite, Magnetite, Limonite, Siderite).
 - Manufacturing process of Cast Iron in a Blast Furnace.
 - Distinguishing parameters of Steel, Wrought Iron, and Cast Iron based on carbon content, ductility, and tensile strength.
 - Non-ferrous structural metals: Aluminum (lightweight, corrosion resistance, high cost, low modulus of elasticity).

Unit 9: Paints & Varnishes — [4.1% - 6.5%] of Total Marks

A. Short Answer Questions (SAOs)

2081 Q8.b (5 Marks):

What are varnishes? Describe the various types of varnishes. [5] *(Part of a long question)*

2080 Q6 (8 Marks):

Explain various ingredients of a good paint. List and explain various types of paints. [8]

2079 Q8.c (5 Marks):

Describe the paints and their ingredients. [5] *(Part of a long question)*

B. Long Answer Questions (LAQs)

No standalone long answer questions have been asked directly from this unit.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Standard descriptive chemistry and product-type classification questions.

- **Direct Repetitions:**
 - **REPEATED: Near Identical** — "Ingredients of a good paint and paint classification" was asked in 2080 Q6 (8 marks) and 2079 Q8.c (5 marks).
- **Key High-Priority Topics:**
 - **Paint Constituents:** Base (white lead, red lead, zinc oxide), Vehicle/Carrier (linseed oil, tung oil), Drier (litharge, lead acetate), Thinner/Solvent (turpentine), and Pigments (colorants).
 - **Types of Paints:** Aluminum paint, Enamel paint, Emulsion paint, Cement paint, Anti-corrosive paint—their composition and selective applications.
 - **Varnishes:** Definition, ingredients (resins, solvents, driers), and classifications (spirit varnish, oil varnish, turpentine varnish, water varnish).

Unit 10: Miscellaneous Materials (Asphalt, Bitumen, Tar, Glass, and Insulating Materials) — [6.5% - 8.9%] of Total Marks

A. Short Answer Questions (SAOs)

2081 Q10.d (4 Marks):

Describe various classification of glass and its properties. [4] (*Part of a long question*)

2081 Q2.b (4 Marks):

Explain in detail about Asphalt, Bitumen and Tar. [4] (*Part of a short question*)

2080 Q11.c (5 Marks):

Electrical, Thermal & Sound Insulating Materials. [5] (*Short note*)

2080 Q10.c (5 Marks):

Describe various classification of glass and its properties. [5] (*Part of a long question*)

2079 Q9.c (5 Marks):

Point out the commonly used insulating materials in electrical industry. [5] (*Part of a long question*)

B. Long Answer Questions (LAQs)

2079 Q11.b (6 Marks):

Draw a comparison between asphalt, bitumen and tar. [6] (*Part of a long question*)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Comparative matrices, material specifications, and engineering use-case matching for glass, binders, and insulations.
- **Direct Repetitions:**
 - **REPEATED: Near Identical** — "Asphalt, Bitumen, and Tar comparison and detailed explanation" was asked in 2081 Q2.b (4 marks) and 2079 Q11.b (6 marks).
 - **REPEATED: Near Identical** — "Glass classification and properties" was asked in 2081 Q10.d (4 marks) and 2080 Q10.c (5 marks).
 - **REPEATED: Similar Concept** — "Insulating materials (electrical/thermal/sound)" was asked in 2080 Q11.c (5 marks) and 2079 Q9.c (5 marks).
- **Key High-Priority Topics:**
 - Comparative distinctions: **Bitumen** (petroleum distillation, soluble in carbon disulfide), **Asphalt** (mixture of bitumen and mineral aggregates), and **Tar** (destructive distillation of coal/wood).
 - **Glass types & engineering properties:** Soda-lime glass, Borosilicate glass, Potash-lime glass, Flint glass, Wired glass, Laminated/Safety glass.
 - **Insulating Materials:** Thermal (glass wool, slag wool, expanded polystyrene), Sound (cork board, acoustic plaster, mineral wool), and Electrical (mica, porcelain, rubber, PVC).

Unit 11: Corrosion & Its Control — [0.0% - 6.5%] of Total Marks

A. Short Answer Questions (SAQs)

2081 Q6 (8 Marks):

What is corrosion? Explain in brief the factors affecting corrosion and its controlling methods. [8]

2080 Q5 (8 Marks):

What is corrosion? What are the factors affecting corrosion and how it can be controlled? [8]

B. Long Answer Questions (LAQs)

No long answer questions have been asked directly from this unit.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This unit is a recent addition or high-focus short question area (8 marks), characterized by a 100% identical question format when it appears.
 - **Direct Repetitions:**
 - **REPEATED: 100% Identical** — "Corrosion definition, factors, and control methods" is 100% identical in 2081 Q6 (8 marks) and 2080 Q5 (8 marks).
 - **Key High-Priority Topics:**
 - Electro-chemical theory of corrosion (anodic and cathodic reactions).
 - **Factors affecting corrosion:** Presence of moisture and oxygen, pH level of environment, chemical impurities in metal, environmental salinity and temperature, stress concentration areas.
 - **Corrosion Control Methods:** Cathodic protection (sacrificial anode, impressed current), protective coatings (galvanizing, tinning, electroplating, painting), alloying (using stainless steel), and environmental modification (inhibitors, dehydration).
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Subject Priority Planner (ROI-Based Study Guide)

The following priority planner ranks all chapters/units from highest return-on-investment (highest exam value for the effort spent) to lowest, based on historical mark percentages and frequency of occurrence in FWU exams.

Rank	Unit No.	Unit Name	Exam Priority Class	Historical Mark Percentage	Core Focus Areas to Pass/Score
1	Unit 6	Concrete & Admixtures	Highest	16.3% - 22.0%	Properties of fresh & hardened concrete, workability testing, concrete manufacturing steps, curing methods, admixtures.
2	Unit 3	Clay Products (Bricks and Tiles)	Highest	9.8% - 13.8%	Traditional brick manufacturing stages, brick quality tests, tiles classification, efflorescence defect & its removal.
3	Unit 5	Cement	High	5.7% - 12.6%	Bogue's compounds, chemical reactions inside the kiln, dry process flowchart, tests (soundness & compressive strength).
4	Unit 8	Metals & Alloys	High	9.8% - 10.6%	Factors affecting steel properties, cast iron production, alloy steel, aluminum advantages/disadvantages, iron ores.
5	Unit 2	Building Stones	High	8.9% - 10.6%	Crushing & water absorption tests, dressing purposes, quarrying location constraints, preservation methods.
6	Unit 7	Timber & Wood Products	High	6.5% - 10.6%	Timber defects (shakes, knots), seasoning methods (air vs. kiln), preservation treatments, timber products (plywood).
7	Unit 10	Miscellaneous Materials	Medium	6.5% - 8.9%	Asphalt vs. Bitumen vs. Tar comparison, glass types & properties, electrical/thermal/sound insulation materials.
8	Unit 1	Intro & Engineering Properties	Medium	6.5% - 6.5%	Porosity, toughness, creep, brittleness, fatigue, resilience definitions, and factors for civil material selection.
9	Unit 4	Lime & Lime Mortar	Medium	3.3% - 7.7%	Chemical Lime Cycle, quick lime vs. hydraulic lime, calcination & slaking, properties of lime mortars.
10	Unit 9	Paints & Varnishes	Low	4.1% - 6.5%	Key paint ingredients (base, vehicle, solvent), paint classifications, varnish types & components.
11	Unit 11	Corrosion & Its Control	Low	0.0% - 6.5%	Electro-chemical corrosion process, factors accelerating rust, and preventive techniques (galvanizing, cathodic protection).